Microbe Perturb-seq – Tables

Companion tables for [[microbe-perturb-seq.method-review-and-costing]]. Generated by `experiments/019-perturb-seq-costing/scripts/render_tables.py` *– do not hand-edit.*

Table 1. mRNA content per cell

The denominator for every capture-efficiency claim. **Rows marked (external) are NOT in the `torchcell-library` mirror** – no mirrored bacterial paper states a total mRNA content, only what its assay detected. Pull the primary paper into the mirror before using these in a proposal.

Organism	Quantity	Value	Source
S. cerevisiae	Total mRNA molecules per cell	**20,000 – 60,000**	Jariani 2020 (`paper.md:31`)
S. cerevisiae	Working figure (Brettner)	~30,000	Brettner 2024 (`paper.md:70`), citing Miura 2008
S. cerevisiae	Implied by 3,000 genes x 3.5 molecules	~10,500	Nadal-Ribelles 2024 (`paper.md:107`)
S. cerevisiae	Total RNA per cell (~85% rRNA)	0.7 – 1 pg	Nadal-Ribelles 2024 (`paper.md:49`)
S. cerevisiae	Protein-coding genes	~6,000	SGD / R64
E. coli	Total mRNA per cell, **rich medium (LB)**	**~7,800**	Bartholomaus 2016, BNID 112795 (external)
E. coli	Total mRNA per cell, **minimal medium**	~2,400	Bartholomaus 2016, BNID 112795 (external)
E. coli	Total mRNA per cell, medium growth rate	~3,000	Taniguchi 2010, *Science* 329:533 (external)
E. coli	rRNA share of all transcripts	**>95%**	Brandner 2025 (`paper.md:71`)
E. coli	Protein-coding genes	~4,400	K-12 MG1655 annotation
Mammalian	Total mRNA molecules per cell	50,000 – 300,000	Jariani 2020 (`paper.md:31`)

> **A yeast cell holds roughly 4x the mRNA of an *E. coli* cell in LB and ~10x one in minimal medium, over ~1.4x the gene count.** That is why the same split-pool chemistry recovers 410 mRNA UMIs/cell in yeast and 235–397 in bacteria, and why the bacterial rRNA burden (>95% vs ~85%) hurts more.

Table 2. Published single-cell RNA-seq runs, yeast and microbial

Study	Platform	Isolation	Cells	mRNA UMIs/cell	Genes/cell	Perturbation readout
Gasch 2017	Fluidigm C1 / SMART-Seq v4	plate	163	735–5,437	–	–
Nadal-Ribelles 2019	FACS + STRT-seq (5' end)	plate	285	3,339	–	–

Study	Platform	Isolation	Cells	mRNA UMIs/cell	Genes/cell	Perturbation readout
Jackson 2020	10x Chromium 3' v2	droplet	38,225	2,000	684	yes
Jariani 2020	10x Chromium 3' v2 + in-droplet zymolyase	droplet	6,118	528–1,261	–	–
Urbonaite 2021	yeastDrop-Seq (table-top)	droplet	844	840–1,336	443–619	–
Dohn 2021 (mDrop-seq)	mDrop-seq	droplet	74,814	–	193–967	–
Boocock 2023	10x Chromium 3' v2 + v3	droplet	100,220	1,514–6,559	–	yes
Brettner 2024 (SPLiT-seq)	yeast-optimized SPLiT-seq	split-pool	43,388	120–700	100–600	–
microSPLiT (B. subtilis)	microSPLiT	split-pool	1,000,000	397	230	–
mapSPLiT (E. coli, CRISPRi/a)	mapSPLiT (microSPLiT + GBC)	split-pool	76,068	325	161	yes

Table 3. Cost per cell, library preparation only

Quoted = the source states a per-cell rate. **Derived** = a total divided by a cell count the same source supplies.

Method	Organism	Isolation	$ / cell	Basis	Source
MATQ-seq	*bacteria*	plate	$50.00	derived	Gaisser 2024, *Nat Protoc*, Suppl. Table 2
Plate FACS + STRT-seq (Nadal-Ribelles 2019)	*S. cerevisiae*	plate	$4.15	quoted	Jariani 2020, *eLife* 9:e55320
10x Chromium 3' v2 (Jariani 2020)	*S. cerevisiae*	droplet	$0.3000	quoted	Jariani 2020, *eLife* 9:e55320
ProBac-seq	*bacteria*	droplet	$0.1649	derived	Gaisser 2024, *Nat Protoc*, Suppl. Table 2
PETRI-seq	*bacteria*	split-pool	$0.0367	derived	Gaisser 2024, *Nat Protoc*, Suppl. Table 2
microSPLiT (1 sublibrary)	*bacteria*	split-pool	$0.0172	derived	Gaisser 2024, *Nat Protoc*, Suppl. Table 2
SPLiT-seq (Brettner 2024)	*S. cerevisiae*	split-pool	$0.00500	derived	Brettner 2024, *Yeast* 41:242

Method	Organism	Isolation	$ / cell	Basis	Source
microSPLiT (full scale)	*bacteria*	split-pool	$0.00251	derived	Gaisser 2024, *Nat Protoc*, Suppl. Table 2

Spread, cheapest to dearest: ~**19,920x**. Every row excludes sequencing – see Table 4.

Table 4. Cost per cell, end to end

6,000 genes at 250 cells/gene. Per cell that survives QC **and** carries an identified guide.

Platform	Reagents $/cell processed	Loaded $/usable cell	Hidden multiplier	Sequencing share
SPLiT-seq (Brettner, as published)	$0.0049	**$0.1044**	21.3x	81.2%
SPLiT-seq + rRNA depletion	$0.0053	**$0.0383**	7.2x	44.2%
10x Chromium X (GEM-X 3')	$0.1076	**$0.2424**	2.3x	19.2%

The reagents-only comparison says split-pool beats droplet by ~22x. End to end it is **2.3x**, and **6.3x** with rRNA depletion.

Table 5. Brettner 2024 SPLiT-seq line items

All from `brettnerUltraHighthroughputMassively2024` Methods 5.1.9 (`paper.md:183`).

Item	Cost	Scaling
IDT barcode plates (RT r1 + ligation r2 + r3)	$7,699.40	**one-time, 215 runs**
Standalone oligos (lyophilized)	$270.60	one time
Maxima H Minus Reverse Transcriptase (ThermoFisher EP0753)	$840.00	**per run**
T4 DNA Ligase (NEB M0202M)	$270.00	**per run**
RNase inhibitors (SUPERase-In + Enzymatics)	$127.00	**per run**
Barcoding oligos drawn from the plates	$36.00	**per run**
Dynabeads MyOne Streptavidin C1	$14.00	per sublibrary
WGS fragmentation + ligation library prep (Enzymatics)	$39.00	per sublibrary

Rolled up: start-up ~**$10,000**, per protocol **$1,300–1,700**, plus **$55** per sublibrary.

Independent replication – Gaisser 2024 microSPLiT, Suppl. Table 1:

	1 sample	24 samples	48 samples
Reagents	$710.24	$1,858.92	$2,929.26
Consumables	$80.54	$156.23	$246.60
Barcodes	$20.67	$20.67	$20.67
Total	**$790.78**	**$2,035.82**	**$3,196.53**

Start-up **$8,142.52**, of which **$7,844.52** is the custom IDT barcode plates – against Brettner's $7,699.40 for the same item, 1.9% apart. Note barcode cost per run is **flat** regardless of sample count.

Table 6. Barcode collision rate

Gaisser's formula $100(1-((B-1)/B)^{C-1})$, $B = 96^R \times S$. **The sublibrary index S is a real barcode dimension** – cells split into sublibraries after the last in-cell ligation.

Rounds	Sublibraries	Barcode space	@ 45k cells	@ 480k cells	@ 3M cells	@ 12M cells
3	1	884,736	4.96%	41.87%	96.63%	100.00%
3	24	21,233,664	0.21%	2.24%	13.18%	43.17%
3	96	84,934,656	0.05%	0.56%	3.47%	13.18%
4	1	84,934,656	0.05%	0.56%	3.47%	13.18%
4	**24**	**2,038,431,744**	0.00%	0.02%	0.15%	0.59%
4	96	8,153,726,976	0.00%	0.01%	0.04%	0.15%

A genome-scale screen barcodes ~12M cells. Three rounds cannot reach 1% at that scale even across 96 sublibraries; four rounds reach 0.59% with 24. The 4th plate costs ~$2,566 once.

Table 7. Cells per perturbation

Two independent constraints: a biological floor of **100 cells** (field standard, stated by Brandner) and a pseudobulk depth target of **200,000 mRNA UMIs**. The larger binds.

Method	mRNA UMIs/cell	Cells/perturbation	Binding constraint	Cells for 6,000 genes
SPLiT-seq as published	410	**488**	sequencing depth	2,928,000
SPLiT-seq + rRNA depletion	861	**233**	sequencing depth	1,398,000
10x v2 (Jariani)	894	**224**	sequencing depth	1,344,000
10x v3 (Jackson)	2,000	**100**	biological floor	600,000
10x v3 (Boocock)	4,036	**100**	biological floor	600,000

Detection ladder – what a pseudobulk budget reaches, given ~16 UMIs of a gene are needed to call a 2-fold change at 80% power:

Tier	Pseudobulk UMIs / perturbation	Lowest detectable gene, 2-fold	… at 1.5-fold
screen (strong coherent effects only)	50,000	9.8 copies/cell	28.6 copies/cell
standard (Brettner's own 500-cell heuristic)	200,000	2.4 copies/cell	7.2 copies/cell
deep (reaches low-abundance regulators)	1,000,000	0.5 copies/cell	1.4 copies/cell

Table 8. Genome-scale CRISPRi perturb-seq budget

6,000 genes, MAGIC-style 6 guides/gene, one environment. UIUC Carver rates effective 2026-08-01. Excludes labour, oligo pool synthesis, strain construction, and the ~$10,000 one-time split-pool start-up.

Cells/gene	Platform	Usable cells	Sequenced	Runs	Read pairs	Reagents	Sequencing	Total
100	SPLiT-seq	600,000	2,400,000	5	50.0e9	$11,470	$50,880	**$62,350**
100	SPLiT-seq + rRNA depl.	600,000	2,400,000	5	10.0e9	$12,470	$12,720	**$25,190**
100	10x Chromium X	600,000	1,090,910	55	27.3e9	$117,865	$28,620	**$146,485**
250	SPLiT-seq	1,500,000	6,000,000	13	125.0e9	$29,470	$127,200	**$156,670**
250	SPLiT-seq + rRNA depl.	1,500,000	6,000,000	13	25.0e9	$32,070	$25,440	**$57,510**
250	10x Chromium X	1,500,000	2,727,273	137	68.2e9	$293,591	$69,960	**$363,551**
500	SPLiT-seq	3,000,000	12,000,000	25	250.0e9	$57,185	$251,220	**$308,405**
500	SPLiT-seq + rRNA depl.	3,000,000	12,000,000	25	50.0e9	$62,185	$50,880	**$113,065**
500	10x Chromium X	3,000,000	5,454,546	273	136.4e9	$585,039	$136,740	**$721,779**

Table 9. UIUC Carver Biotechnology Center sequencing rates

Effective 2026-08-01, retrieved 2026-08-18. On-campus rate; external users pay +30.8%. **The core has no NextSeq and no NovaSeq 6000.**

Option	Instrument	$ / lane	Read pairs / lane	$ per M read pairs
Nova X 25B PE150, 8+ lanes	NovaSeq X Plus	$3,180	3,200,000,000	$0.99
Nova X 25B PE150, per lane	NovaSeq X Plus	$3,380	3,200,000,000	$1.06
Nova X 10B PE150, 8+ lanes	NovaSeq X Plus	$1,710	1,200,000,000	$1.43
Nova X 10B PE150, per lane	NovaSeq X Plus	$2,000	1,200,000,000	$1.67
Nova X 1.5B SR100, per lane	NovaSeq X Plus	$1,360	800,000,000	$1.70
Nova X 10B PE150, SPLIT LANE	NovaSeq X Plus	$1,160	600,000,000	$1.93
Nova X 1.5B PE150, per lane	NovaSeq X Plus	$1,580	800,000,000	$1.98
i100 25M SR100nt	MiSeq i100	$986	25,000,000	$39.44
i100 25M 2x150nt	MiSeq i100	$1,450	25,000,000	$58.00
i100 titration run (26nt + indexes only)	MiSeq i100	$398	5,000,000	$79.60
i100 5M 2x150nt	MiSeq i100	$661	5,000,000	$132.20

Both split-pool and droplet require **paired-end**; neither can use a single-read flow cell. That costs nothing here – the 25B PE150 8+ lane rate ($0.99/M pairs) beats the 10B SR100 rate ($1.06/M).

Table 10. Multiplexing

Target is 250-cells/gene main-effect precision over 6,000 genes.

Plex k	Cells for main effects	Cells per *named* pair (6,000 genes)	… in a 200-gene sublibrary
1	1,500,000	0.000	0.0
2	750,000	0.042	37.7
3	500,000	0.083	75.4
5	300,000	0.167	150.8
10	150,000	0.375	339.2

Main effects get k times cheaper. Named pairwise interactions never become reachable genome-wide (1.8e7 pairs), but a focused 200-gene sublibrary at $k = 3$ puts ~75 cells on each of its 19,900 pairs.